Income Effects in Labor Supply: Evidence from Child-Related Tax Benefits

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Abstract

A parent whose child is born in December can claim child-related tax benefits when she files her tax return a few months later. Parents of children born in January must wait more than a year before they can receive child-related tax benefits. As a result, families with December births have higher after-tax income in the first year of a child's life than otherwise similar families with January births. This paper estimates the corresponding income effect on maternal labor supply, testing whether mothers who give birth in December work and earn less in the months following birth. We use data from the American Community Survey, the Survey of Income and Program Participation, and the 2000 Decennial Census. We find that December mothers have a lower probability of working, particularly in the third month after a child's birth. Earnings data from the SIPP indicate that an additional dollar of child-related tax benefits reduces annual maternal earnings in the year following a child's birth by approximately one dollar.

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The birth of a child, at any point during a year, reduces her parents' tax liability for that year. This creates a discontinuity in the tax treatment of children born around the turn of the year. A child born on December 31 can be claimed on that year's tax return, typically filed in the early months of the following calendar year. As a result of the December birth, this child's parents will have lower tax liability, and will typically receive a larger tax refund, in the first few months of the child's life. With a January 1 birth, parents must wait for more than a year before they receive the tax benefits associated with claiming the child. This paper uses the discontinuous tax treatment of children born around the turn of the year to identify income effects on maternal labor supply, testing whether December mothers work and earn less during a child's first year of life as a result of the tax benefits they receive.

Tax policy changes have often been used to estimate the elasticity of taxable income and of labor supply. However, many tax changes affect both a filer's marginal tax rate and after-tax income, making it difficult to separately identify the substitution effect and income effect. While the tax treatment of newborns generates differences in after-tax income of December and January parents in the first year of a child's life, both sets of parents face the same schedule of marginal tax rates on income earned after the child's birth. This situation provides an opportunity for a well-identified estimate of the income effect on labor supply.

Implementing this empirical test requires a dataset with precise information on date of birth, and with information on labor supply and earnings measured at various points after a mother gives birth. No single dataset is ideal for our purposes, and in this paper we present related evidence from three datasets, each with particular advantages and disadvantages. We use data from the restricted-access version of the 1999 through 2008 rounds of the American Community Survey (ACS); restricted-access data from the 2000 Decennial Census; and publicly available data from the 1996, 2001, and 2004 panels of the Survey of Income and Program Participation (SIPP). The key advantage of the ACS and Decennial Census is that the data include exact date of birth. This allows us to identify mothers giving birth in narrow windows around the turn of the year. We also know exactly how much time

has elapsed between when a mother gives birth and when she reports employment status and earnings. This is important because the time surrounding the birth of a child is, on average, characterized by rapid changes in employment (Laughlin 2011). Neither the ACS nor the Decennial Census has a longitudinal component. The SIPP data offer the important advantage of following mothers over time. Because we can compare a mother's pre-birth and post-birth earnings in the SIPP, these data allow us to better address concerns that mothers giving birth in December and January might have systematically different attachment to the labor force even before receiving different tax treatment.

We find that mothers receiving more generous tax treatment, by virtue of a December rather than January birth, reduce their labor supply and earnings in the year following a child's birth. Data describing contemporaneous labor supply measures in the ACS and SIPP show very similar patterns. In both datasets we find that December mothers are about 6 percentage points less likely to be currently working in the third month after birth. As this corresponds to the time at which many maternity leaves come to an end, it is a plausible moment at which additional cash might have a strong impact on labor supply decisions.

Earnings data from the SIPP suggest that every additional dollar of tax benefit reduces earnings in the year after birth by about a dollar. Because earnings are reported at monthly frequency in the SIPP, we can see that the fourth and fifth months after birth are particularly important in creating this dollar-for-dollar reduction in earnings, although there are sustained differences in earnings throughout the first year following birth. Data from the ACS and Decennial Census do not show a statistically significant relationship between tax benefits and earnings after birth. We argue that the longer retrospective period over which earnings are reported in the ACS and Decennial Census as well as the specific time structure of reported earnings make it more difficult to statistically distinguish this effect. Although consensus estimates of the income elasticity of female labor supply are typically small (Blundell and MaCurdy 1999), the year after a child's birth could plausibly be a time at which mothers' labor supply decisions are particularly sensitive to non-wage income.

The assertion that any difference in the labor supply of December and January mothers measures an income effect relies on the assumption that the two groups of mothers are similar, except for their tax treatment. This assumption will be violated if mothers manipulate birth timing around the turn of the year in response to tax benefits. In fact, previous research finds that the probability of a birth in the last week of December, relative to the first week of January, is strongly increasing with the value of child-related tax benefits (Dickert-Conlin and Chandra 1999). This result is based on observations from years 1979 to 1993. More recent estimates of the tax sensitivity of birth timing are much smaller (Wingender 2010, Schulkind and Shapiro 2014, LaLumia, Sallee and Turner 2015). Aggregate birth counts from Vital Statistics data show that the distribution of births across December and January has remained steady over time, even as the average value of child-related tax benefits has grown. Our analysis of ACS data shows no statistically significant relationship between the size of tax benefits and the probability of a December birth for the main analysis sample, consisting of births occurring in the last two weeks of December and first two weeks of January. There is evidence of a small positive relationship when the sample is restricted to births occurring at most one week before or after the turn of the year. Furthermore, we show that many of the mean characteristics of December and January mothers are statistically indistinguishable. We use this evidence to argue that our results can be interpreted as a response to differences in tax treatment. However, we acknowledge the potential for manipulation of birth timing to be influencing our results. We present additional discussion of potential seasonal variation in birth counts and maternal characteristics in Section 3.

The paper proceeds in the following way. Section 1 summarizes previous estimates of income effects on female labor supply and offers an overview of the major child-related tax benefits. Section 2 provides detail on the three datasets used in the analysis. The validity of the paper's identifying assumption, that the labor supply of December and January mothers would evolve similarly absent any difference in tax treatment, is probed in section 3. Section 4 outlines the empirical methods used, section 5 presents results, and section 6 concludes.

1 Literature Review

1.1 Maternal Labor Supply

This paper adds to the literature estimating the elasticity of female labor supply. A comprehensive survey of this work is provided by Blundell and MaCurdy (1999). Many estimates are of the uncompensated labor supply elasticity, which reflects both income and substitution effects. Among the papers cited by Blundell and MaCurdy (1999), most estimates of wives' uncompensated wage elasticity are in the range of 0.5 to 1, with a median of 0.78, and most estimates of the income elasticity are in the range of -0.05 to -0.4.[1] All of these estimates predate the 1999-2008 period analyzed in this paper, and there is evidence that female labor supply elasticities, both uncompensated and compensated, have fallen over time. Blau and Kahn (2007) estimate that wives' labor supply elasticities with respect to non-wage income are very small over the entire 1980-2000 time period, but particularly close to zero by the last years analyzed. In each time period, they find slightly larger (in absolute value) elasticities with respect to non-wage income for wives with children under age 6. Similarly, Heim (2007) finds that wives' labor supply elasticities have fallen over time.

The income elasticity of female labor supply may be quite different in the months shortly after a child's birth than at other times. One reason to expect a difference is that the marginal utility of time at home is likely particularly high when an infant is first born, before gradually falling as the child grows (Klerman and Leibowitz 1994). A second reason is that maternity leave policies explicitly provide job protection for new mothers. Outside of maternity leave, responding to a positive non-wage income shock by temporarily reducing labor supply could result in job loss. However, the federal Family and Medical Leave Act (FMLA) mandates that eligible parents can take up to 12 weeks off of work to care for a newborn, while retaining the right to return to the same or an equivalent job (Ruhm 1997).

[1] Estimates of the income effect on labor supply that rely on a very different source of exogenous variation in income, lottery winnings, find an income elasticity of -0.11, with no significant difference between men and women (Imbens, Rubin and Sacerdote 2001).

Some states offer mothers longer periods of entitlement to unpaid leave. Evidence shows that the legal protection provided by mandated unpaid leave is associated with increased probabilities of mothers being on leave shortly after birth (Han, Ruhm and Waldfogel 2009).

Evidence from the introduction of *paid* maternity leave policies shows that this source of additional income lengthens maternity leaves. Rossin-Slater, Ruhm and Waldfogel (2011) study California's 2004 introduction of a paid leave benefit equivalent to 55% of pre-leave wages, up to a cap of approximately $1000 per week. This program increased average maternity leaves by about three weeks, from an average of three weeks to six weeks, with the largest effects estimated for relatively disadvantaged mothers. If paid maternity leave lengthens mothers' time away from work, it seems reasonable that income from other sources, including tax benefits, could also reduce labor supply following a child's birth. The most direct evidence on how government-provided cash benefits for families with children affect labor supply of new mothers comes from policy changes in Spain (Gonzalez 2013) and Canada (Milligan and Stabile 2009). Gonzalez (2013) uses regression discontinuity techniques to compare mothers giving birth just before and after the unanticipated introduction of a substantial universal child benefit.[2] Eligible mothers were four to six percentage points less likely to be working one year later. Milligan and Stabile (2009) find that a more generous child benefit for Manitoba welfare recipients with children ages 0 to 5 is associated with reductions in work among low-income families. While these results suggest a negative income effect of U.S. tax benefits on maternal labor supply, it is difficult to extrapolate across countries with different parental leave policies and with differing levels of publicly-provided childcare.

The welfare consequences of delayed return to work following the birth of a child are not clear. Some evidence suggests that longer maternity leaves are associated with positive outcomes: improved infant health (Rossin 2011), increased rates of breastfeeding and infant

[2]The one-time benefit payment is €2500, about 4.5 times the monthly earnings of a full-time minimum wage worker. For comparison, the average child-related tax savings in our ACS sample is about $1200, or about 1.1 times the monthly earnings of a full-time worker receiving the $6.55 minimum wage in place at the end of our sample period.

immunization (Berger, Hill and Waldfogel 2005), and reductions in the number of depressive symptoms displayed by new mothers (Chatterji and Markowitz 2005). On the other hand, studies analyzing expanded maternity leave in Canada find no significant positive effect on outcomes of young children (Baker and Milligan 2010, Baker and Milligan 2015).

Outside of the specific case of maternity leave benefits, the welfare implications of any transfer that affects labor supply will depend on the extent to which the overall labor supply effect is composed of income and substitution effects. Chetty (2008) shows that optimal unemployment insurance (UI) benefits are larger when UI benefits primarily affect unemployment duration through a non-distorting income (liquidity) effect rather than through a distortionary substitution effect. Our finding of a substantial income effect for new mothers suggests that larger benefits for this group would generate little deadweight loss.

1.2 Child-Related Tax Provisions

Maag (2013) describes the elements of the tax code that provide the most substantial aid to filers with children. The Earned Income Tax Credit (EITC) is a refundable credit calculated as a function of earnings and targeted towards low- and middle-income workers. It is not exclusively for parents but is typically much larger for parents. Using aggregate data for 2008, the average credit for an EITC recipient with no qualifying children was $252, the average credit for an EITC recipient with one qualifying children was $1996, and the average credit for an EITC recipient with two or more qualifying children was $3105 (Tax Policy Center 2011). Over 90% of all EITC payments go to families in the bottom two quintiles of the income distribution. The Child Tax Credit (CTC), up to $1000 per child, is not fully refundable and hence is of less benefit to low-income families. It is phased-out at higher incomes. Families in the middle three quintiles each receive approximately 25% of the benefit from the CTC. The dependent exemption, which ranged from $2750 in 1999 to $3500 in 2008, is available to all but the highest-income families. As an exemption, its value is higher for filers facing high marginal tax rates. Head of household filing status allows an

unmarried parent claiming her first child to move to a tax schedule with wider brackets and a larger standard deduction than the schedule for single filers.

All four of the provisions described above are included in our calculations of child-related tax benefits.[3] In aggregate, the real federal tax savings associated with child-related provisions totaled approximately $105 billion in 2000 and $145 billion in 2010 (Isaacs, Hahn, Rennane, Steuerle and Vericker 2011). The single most costly provision is the EITC, followed by the Child Tax Credit and the dependent exemption. Refundable portions of the tax credits account for most of the growth over the decade.

Other authors have used variation in child-related tax benefits to estimate parameters other than income elasticities. Looney and Singhal (2006) and Dokko (2008) consider the anticipated change in marginal tax rates experienced by some parents when a child turns 19 and ages out of eligibility for the dependent exemption. Only those parents whose income is very close to the edge of a tax bracket are pushed into a different bracket, with a different marginal tax rate, by losing a dependent exemption. Both papers find that parental labor supply responds to the resulting change in marginal tax rate, estimating large intertemporal substitution elasticities. Feldman, Kawano and Katuscak (2014) consider the predictable, lump-sum change in tax liability associated with aging out of eligibility for the CTC at age 17 in order to estimate taxpayers' understanding of the tax code. A fully informed taxpayer should anticipate this loss and make no adjustment to labor supply at the time of the loss. Taxpayers who learn about the loss of credit ex post either understand that losing a lump sum credit has no effect on marginal tax rates, and do not reduce labor supply, or are confused about the nature of the tax change. The authors find a reduction in labor income among families losing the credit, interpreting this as evidence that taxpayers incorrectly attribute the increase in tax liability to an increase in marginal tax rates.

[3]We do not include the Child and Dependent Care Tax Credit (CDCTC), which reimburses parents for some portion of out-of-pocket child care expenses paid in order to allow a parent to work. The reason for excluding this from our calculation is that we do not observe child care expenses in any of our datasets. The total tax savings delivered to families through the CDCTC is only about one-tenth of the tax savings associated with the dependent exemption, and about one-twentieth of the tax savings associated with the EITC, so our inability to calculate this benefit likely has little effect on our results.

2 Data

This paper makes use of data from the ACS, the SIPP, and the Decennial Census. Here we describe the samples drawn from each of these datasets, as well as variables that are particularly important for our analysis.

2.1 ACS

We use data from the 1999 through 2008 rounds of the ACS. This is a continuously-fielded cross-sectional survey. It was introduced as the Decennial Census long form was being phased out, and covers many of the same topics. Respondents are asked about their employment status at the time of the interview, and about their income over the 12-month period prior to the interview.

In the restricted-access ACS files, exact date of birth is recorded for all household members. The exact date of the ACS interview is also available. We use these dates to calculate the time elapsed between a child's birth and the ACS interview. We restrict attention to cases in which a child was born no more than one year prior to the interview. Using information on relationships between household members, we match children to their mothers. If a mother is matched to more than one child born within the year prior to the interview date, we use information on the youngest child to compute the time elapsed since the mother's most recent birth. We restrict the sample to mothers whose most recent birth took place in the four weeks spanning the turn of the year, stretching from December 18 to January 14. We further restrict attention to mothers who are between the ages of 15 and 50 and who have had some recent attachment to the labor force, measured as having worked in the previous five years. This process results in a sample of 16,384 mothers, with approximately equal numbers giving birth in late December and in early January.

We calculate the real tax savings a mother would have realized if her child had been born in December using the NBER's TAXSIM program.[4] We estimate tax liabilities in two

[4]We are grateful to Dan Feenberg for providing a version of TAXSIM for use in the RDC.

different scenarios. First, we use household composition as reported at the time the mother is interviewed. Second, we remove the newborn child (or children, in the case of twins). For all mothers, the removal of a child from the return changes the number of dependent exemptions she can claim. Depending on income level, it may change the amounts of EITC and CTC for which she is eligible. If an unmarried mother has no other children living with her, this removal of children also causes her filing status to change from head of household to single. We compute the difference between the tax liability when the newborn child is excluded from the household (the higher of the two tax values) and the tax liability when the newborn is included, and refer to this difference as the child tax benefit. TAXSIM incorporates all relevant elements of federal and state tax codes, including state-specific EITC programs, and our measure of tax savings accounts for both federal and state taxes. We assume that taxpayers take the standard deduction rather than itemizing deductions.[5]

The TAXSIM calculations use information on several types of income. Ideally, we would like to measure income received during a tax year, either the year in which the birth occurred (in the case of December births) or the year just prior to the birth (in the case of January births). However, the reference period for ACS respondents does not correspond to a tax year. Instead, ACS respondents report income for the 12 months prior to the interview date. The difference between our preferred reference period and the actual reference period is unlikely to affect types of income that are independent of female labor supply. However, if December and January mothers have different patterns of labor supply after a child's birth, they may report systematically different labor income amounts and thus have different calculated child-related tax benefits. This problem will not be particularly severe for mothers whose interview date occurs very close to the date at which they gave birth, but will likely be larger for mothers interviewed several months after birth.

[5]This assumption is likely correct for the majority of our sample, as the percentage of all returns taking the standard deduction ranged between 62.6% and 67.5% over the time period of our analysis (Statistics of Income 2015). To the extent that this assumption is incorrect, though, it will cause us to overstate the value of child-related tax benefits. Presumably those who choose to itemize do so because it lowers their tax liability, which in turn will reduce the value of nonrefundable tax credits, including the Child Tax Credit.

To address this problem, we compute one version of the tax benefit variable in which all components of income are taken directly from respondents' self-reports, and another in which predicted labor income replaces self-reported labor income. The income components always taken directly from respondents' reports are property income (interest, dividend, and rental income) and, for married women, husbands' earnings.[6] To predict a mother's wage earnings and, if she reports being self-employed, her business earnings we use a sample of mothers whose youngest child is at least one year and no more than two years old at the time of the ACS interview. We regress the earnings of these mothers on demographic characteristics, and use the resulting coefficients to predict earnings for the mothers of younger children in our main sample. The mothers used in the earnings prediction equation should be similar to mothers of slightly younger children, but for them the 12-month reference period for reported income does not include the time at which they gave birth.

2.2 SIPP

We use data from the 1996, 2001, and 2004 panels of the SIPP. Each panel is a nationally representative longitudinal survey that follows households for three to four years. Households are interviewed every four months, and respondents provide information on monthly employment status and income.

To construct the sample, we identify all children born during a SIPP panel or up to twelve months before the start of the panel, and with a birth month of December or January. Because exact date of birth is not reported in the public-use SIPP, we cannot restrict attention to those with birth dates in the last two weeks of December or the first two weeks of January. Children are linked to their mothers, and mothers are linked to their spouses. Only households in which a mother and child can be linked are included in the analysis. We keep mothers who are between the ages of 15 and 50 and who have had some recent attachment

[6] If a mother is living with either a husband or male partner, we control for his income in the regression analysis. However, following the provisions of the tax code, his income is used in calculating child-related tax benefits only in cases where the couple is married.

to the labor force, as indicated by working at least one week in the twelve months prior to birth. After applying all restrictions, we have a sample of 1051 mothers.

As we did for the ACS sample, we calculate the potential tax savings associated with a December birth for every mother in the SIPP sample. We use reported income amounts from pre-birth months as inputs to the TAXSIM calculation. We sum monthly income amounts to an annual level, scaling up in cases where we observe fewer than twelve months of pre-birth income. We use information on wage income, property income, dividends, pensions, and unemployment benefits. We assume that married couples file jointly, that unmarried mothers use head of household filing status, and that filers take the standard deduction.

2.3 Decennial Census

We use data from the 2000 Decennial Census, conducted in April of 2000. Exact date of birth is reported for every household member. We identify infants born between December 18, 1998 and January 14, 1999 as well as infants born between December 18, 1999 and January 14, 2000. We link these infants to their mothers using information on household relationships. We restrict the sample to mothers linked to children born in the windows of interest, who are between the ages of 15 and 50, and who have worked at some point in the five years prior to the 2000 Decennial Census.

Adult respondents reported annual income earned during calendar year 1999. We consider two groups of mothers. The first group consists of 33,397 mothers who gave birth between December 18, 1998 and January 14, 1999. These mothers report earnings from a period that is almost entirely post-birth. The entire earnings reference period is post-birth for the December 1998 mothers. At most two weeks of the earnings reference period is pre-birth for the January 1999 mothers. We use this group to test the hypothesis that December mothers will earn less in the year after a child's birth than similar January mothers because of the more generous tax treatment they face. Additionally, we consider a second group of 33,860 mothers who gave birth between December 18, 1999 and January 14, 2000. The 1999 wage

income reported by these mothers is entirely pre-birth for the January mothers, and includes at most two weeks of post-birth earnings for the December mothers. We use this group of mothers to test the identifying assumption that December and January mothers would have similar labor supply and earnings absent any difference in tax treatment.

Decennial Census respondents report current employment status, but this information cannot be used to test for differences in contemporaneous labor supply. Decennial Census interviews happen at the same time for all respondents. For mothers who gave birth in December 1999, we observe employment four months after giving birth. For mothers who gave birth in January 2000, we observe employment three months after giving birth. The rapid change in the probability of being currently at work between the third and fourth month after giving birth makes it necessary to compare mothers observed at the same interval after giving birth.

3 Identifying Assumption

The identifying assumption made in this paper is that December and January mothers are similar except for the tax treatment they receive, and that if the two groups of mothers faced the same tax treatment then their post-birth labor supply would evolve similarly. If this assumption holds, the greater after-tax income enjoyed by December mothers in the year after giving birth is exogenous to their previous labor supply behavior and other choices. In this scenario, the exogenous variation in income is what permits a well-identified estimate of the income effect on labor supply.

This assumption could be violated if mothers giving birth at different points in time have different characteristics. Differences in labor force attachment would be particularly problematic for our empirical strategy. While there is seasonal variation in maternal characteristics over the full course of a year, the characteristics that can be measured in Vital Statistics data drawn from birth certificates are similar for December and January mothers

(Buckles and Hungerman 2013). In our ACS and SIPP samples, the observable characteristics of mothers giving birth in late December and in early January are similar, with differences in means generally not statistically different from zero. Summary statistics are shown in Table 1, with ACS data described in Panel A and SIPP data described in Panel B. In the ACS sample of December and January mothers, approximately 74% of each group is white, 71% of mothers are married, and average age is 29 years. About 20% of these mothers have completed some college and an additional 44% have obtained college degrees. The difference in means across December and January mothers is not statistically significant for any of the variables shown in Panel A. Panel B shows similar means for the SIPP sample. There is a great deal of similarity in demographic characteristics across the two samples. Mothers in the SIPP sample are about 28 years old on average, 74% are married, and the mean number of children is about two. Again there are no statistically significant differences between December and January mothers.

Another potential threat to our identification strategy is that seasonal patterns in labor demand may affect December and January mothers differently. Employment and output are regularly higher in the fourth quarter of the year than in the first quarter of the year (Barsky and Miron 1989), and this pattern is particularly strong in the retail sector (Krane and Wascher 1999).[7] It may be the case that in the high-demand month of December, expectant mothers work just up until giving birth. In contrast, in the lower-demand month of January, it may be easier for women to begin their maternity leaves more in advance of giving birth. If December mothers are commencing their leaves closer to the date of delivery, and if December and January mothers take leaves of equal average length, this could result in a lower probability of observing a December mother back at work in the third month after giving birth. It is worth noting that only a minority of working expectant mothers begin their maternity leave much in advance of giving birth. Guendelman, Pearl, Graham, Hubbard, Hosang and Kharrazi (2009) report that only 28% of employed women with a

[7]Seasonal patterns are also strong in the construction industry, which employs few women, and in state and local government, where activity declines in the summer months when many schools are not in session.

first birth in 2001-2003 took antenatal leave. Among a smaller sample of mothers employed full-time as of week 36-39 of pregnancy, only 62 out of 447, or 14%, took any antenatal leave.

Perhaps the strongest threat to our identification is that mothers with the largest tax benefit may shift births from early January to late December through use of a scheduled C-section or induction. If this behavior is occurring, our treatment of child-related tax benefits as an exogenous shock to income is problematic. Aggregate birth counts by exact date of birth, published by the National Center for Vital Statistics, are informative about the potential magnitude of this problem. Figure 1 plots the number of births occurring on each day of December and January, beginning in December 1997 and continuing until January 2003.[8] The more common it is for mothers to shift births into December to accelerate the receipt of tax benefits, the larger would be December's share of births. Reassuringly, births are relatively evenly distributed across December and January. Considering only births in the last week of December and first week of January, 50.5% of births occur in December. December's share is only slightly smaller, 50.4%, when considering either births occurring within two weeks of the turn of the year or when considering the two full months.

Additional evidence on the extent to which tax-motivated shifting of birth timing compromises our identifying assumption is shown in Table 1. The bottom rows of each panel report the average amount of tax savings associated with a December birth for mothers in each group. On average, mothers in the ACS sample receive about $1205 in child-related tax benefits, when benefits are computed as a function of reported income, and about $1300 when predicted labor income replaces reported labor income. Late-December mothers receive about $10 more in child-related tax benefits than early-January mothers would have realized, had their children been born earlier. We see this pattern if tax benefits are computed

[8]This is the full set of years that both overlaps with the ACS and SIPP data used in this paper and for which publicly-available counts by exact date of birth are available. The NCVS has not published birth counts by exact date of birth for years after 2003. In the set of six December-January pairs included in the figure, there is no Saturday observation for December 3, 10, 17, 24, and 31 or January 7, 14, 21, and 28. There is no Sunday observation for December 4, 11, 18, and 25 or January 1, 8, 15, 22, and 29. Daily birth counts are consistently lower on weekends than on weekdays. As a result, we observe higher birth counts on dates that are missing either a Saturday or Sunday over the period plotted in the figure.

as a function of reported income or of predicted income. The direction of this difference is consistent with the possibility of tax-motivated shifting of birth timing, although the difference is not statistically different from zero. Because of the SIPP's longitudinal structure, income from months before a child's birth can be observed, and the potential tax savings associated with a December birth can be computed as a function of pre-birth income. The average amount of this benefit for all mothers in the SIPP sample is $978, with no statistical difference in the benefit amount for December and January mothers. The average dollar amount of tax benefit is smaller for the SIPP sample than for the ACS sample in part because an earlier set of years is included in the SIPP sample, and legislative changes have made child-related tax benefits more generous in more recent years.

We find it reassuring that average tax benefits are statistically indistinguishable for the December and January mothers. To further investigate the potential impact of tax-motivated shifting of birth dates on our main analysis, we estimate equations predicting December birth as a function of tax benefits. Using ACS data, we regress a December birth dummy on the real dollar value of child-related tax benefits (computed as a function of predicted wage income) as well as controls for maternal age, race, education, marital status, and a dummy for being the first child.[9] If larger tax benefits cause women to shift births to late December from early January, the coefficient on the tax benefit term will be positive. Results are shown in Table 2. The first column includes our main analysis sample, consisting of mothers with births in the last two weeks of December and first two weeks of January. In this sample, a larger tax benefit does not have a statistically significant effect on the probability that a child is born in December. The second column of the table narrows the sample to births taking place within a two-week window, including only the last week of December and the first week of January. In this narrower window, which matches the set of birthdays used in Dickert-Conlin and Chandra (1999) and LaLumia et al. (2015), there is evidence that larger tax values are associated with a higher probability of a December birth. An additional

[9]The survey does not ask about all previous children born to a mother. We characterize a child as first-born if a mother has no other own children living in her household at the time of the ACS interview.

$1000 of child-related tax benefits is associated with a 1.7 percentage point increase in the probability of a late-December vs. early-January birth. We view this as a reason to rely on the sample of births occurring in the wider December 18 to January 14 window. The longer time period does not eliminate the potential for strategic manipulation of birth timing to be influencing our results, but it does likely reduce the percentage of mothers in our sample who have intentionally shifted a birth forward in time.

Finally, we provide additional tests of the validity of our identifying assumption using Decennial Census data. For the mothers giving birth in December 1999 and January 2000, who are reporting pre-birth earnings in the 2000 Decennial Census, we expect to see no discontinuous difference in reported earnings on either side of the new year. Figure 2 plots average annual earnings for mothers giving birth on each day between December 1, 1999 and January 31, 2000.[10] Earnings appear to evolve in a smooth fashion through the turn of the year, offering support for our identifying assumption that the earnings of December and January mothers are similar before these mothers are exposed to differing tax treatment.[11]

4 Empirical Strategy

To investigate whether the child-related tax benefits received by December mothers shortly after giving birth reduce their labor supply and earnings, we estimate both OLS and IV regressions. The OLS specification estimates the reduced-form effect of having a child in late December relative to early January. These regressions, which are estimated using ACS and SIPP data, take the form:

$$Y_i = \sum_{k=1}^{T-1} \alpha_k MonthsElapsed{=}k_i + \sum_{k=1}^{T} \beta_k (MonthsElapsed{=}k_i \cdot DecBirth_i) + \gamma X_i + \epsilon_i \quad (1)$$

[10] We cannot produce similar figures for our ACS sample, as the number of mothers giving birth on a particular calendar date, and observed t months after birth, is so small as to generate disclosure risk.

[11] In results not shown, we have also checked the robustness of the SIPP results by using inverse probability weights to address selection of women with different observable characteristics into the group of December mothers. The pattern of results is very similar to what we find without adjustment. Full results are available upon request.

where the dependent variable is a measure of labor supply and the vector X controls for mothers' demographic characteristics such as age, education, marital status, and race.

Regardless of tax treatment, the birth of a child is associated with substantial short-run changes in labor supply. Mothers are very unlikely to be working just after giving birth. The probability of being at work remains low for a few months and then increases, rapidly at first and then more slowly. Because of this strong temporal pattern in labor supply following the birth of a child, Equation 1 controls for the number of months elapsed between the date of a mother's most recent birth and the interview date. This set of controls is represented by the vector of *MonthsElapsed* dummy variables, with the excluded category being twelve months following birth. Reductions in labor supply in the month immediately after birth will be represented by negative coefficients when $k = 1$. As labor supply gradually increases to pre-birth levels, the coefficients should become less negative for larger values of k.

The variables of greatest interest in Equation 1 are the interaction terms *MonthsElapsed* · *DecBirth*. The coefficients on these terms measure any difference in labor supply of December and January mothers, allowing this difference to vary with the length of time elapsed between the child's birth and the time at which labor supply is measured. If more generous tax treatment reduces the relative labor supply of December mothers in a particular month, the coefficient on the corresponding interaction term will be negative.

There are a few differences involved in estimating Equation 1 using cross-sectional ACS data and longitudinal SIPP data. Equation 1 corresponds to the ACS sample, with no t subscripts on the *MonthsElapsed* terms. Each mother is observed in only one of the twelve months after giving birth. In contrast, each mother is observed multiple times in SIPP data. If written to correspond to the SIPP sample, the dependent variable and the *MonthsElapsed* terms in Equation 1 would have t subscripts. In the SIPP sample, the dependent variable is defined as the change in monthly earnings relative to a mother's average monthly earnings calculated over the six months prior to giving birth. The vector X_i is expanded to include the mother's average monthly earnings in the six months prior to birth.

If there are systematic differences in the earnings of December and January mothers even absent any differential tax treatment, the use of first-differenced earnings data and the inclusion of these additional variables should control for the difference. The definition of *MonthsElapsed* is slightly different in the two samples. In the SIPP, it is a function of the calendar month of birth and the calendar month of the reference period. Given that the ACS reports exact date of birth and exact date of interview, we instead define the number of months elapsed as the number of 30-day periods elapsed. In the SIPP analysis, we cluster standard errors at the mother level to account for any serial correlation in the error term.

We consider several dependent variables. The first, available in both the ACS and the SIPP, is a dummy equal to one if a mother is employed and working at the time of the interview. When this is the dependent variable, ACS regressions include all mothers interviewed up to one year after giving birth. A value of zero indicates either that a woman is on temporary leave from a job or that she is out of the labor force. The second dependent variable, available in the ACS, is a dummy equal to one if a mother is employed but is currently not working. Regressions with this as the dependent variable are estimated only for employed mothers, and a value of zero indicates that a woman is working at the time of the interview as opposed to being temporarily absent from an ongoing job.

The third dependent variable that we use is wage and salary income. This is measured over the preceding 12 months in the ACS, as the monthly change relative to pre-birth earnings in the SIPP, and in calendar year 1999 in the Decennial Census. For the ACS mothers in our sample, all of whom are interviewed within one year of having given birth, reported wage income will include a mixture of pre-birth and post-birth earnings. For mothers interviewed very shortly after a child's birth, the window of time covered by the wage question falls almost entirely before the birth, while mothers interviewed twelve months after a child's birth will be reporting entirely post-birth earnings.

Estimating the effect of a December birth on earnings in either the ACS or SIPP can be carried out using Equation 1, where the dependent variable is 12-month earnings in the

ACS and change in monthly earnings in the SIPP. It is just as important to include the *MonthsElapsed* dummies in the wage regressions as in the contemporaneous employment status regressions, but the expected pattern of coefficients is somewhat different. We expect positive coefficients for particularly low values of k in the ACS sample, as the mothers interviewed very shortly after giving birth will be including many months of (likely higher) pre-birth earnings in their 12-month wage measures. We expect these coefficients to decline as k increases, as the time period over which wage income is summed includes more months of post-birth earnings. In the SIPP-based analysis of wage income, we set the dependent variable equal to the difference between current monthly earnings and the mother's own average earnings in the six months prior to birth. For comparability, we include a similar set of *MonthsElapsed* dummies in SIPP wage income regressions. Because wage income is measured at the monthly level, though, we expect a different temporal pattern in the coefficients on these terms. We expect the most negative coefficients in months when maternal labor supply is lowest, very shortly after birth. We expect the coefficients on *MonthsElapsed* terms to become gradually less negative as k increases.

To estimate the effect of a December birth on annual earnings in the Decennial Census, we use the following OLS regression:

$$Annual\ Wage\ Income_i = \beta DecBirth_i + \gamma X_i + \epsilon_i \tag{2}$$

For mothers who gave birth in December of 1998, the Decennial Census wage measure includes post-birth earnings exclusively. For mothers who gave birth in the first two weeks of January of 1999, up to two weeks of pre-birth earnings are included. This can complicate the interpretation of Census results. Under the hypothesis that the higher after-tax income received by December mothers, due to their more generous tax treatment, reduces their labor supply, we would expect a negative coefficient on *DecBirth* when estimating equation 2. However, suppose that all mothers take exactly the same number of weeks of unpaid

maternity leave before returning to work, and suppose that this maternity leave commences on the date of the child's birth. All of the time on unpaid maternity leave will be included in the Census wage measure for January mothers. For December mothers, however, some portion of this maternity leave will have occurred before calendar year 1999 and will not be reflected in 1999 calendar year wages. This will cause 1999 wage income to be systematically higher for December mothers than for January mothers, and will bias upwards the coefficient on *DecBirth* in Equation 2. This problem is mitigated somewhat by using a sample of women who give birth within a narrow window of time, the last two weeks of December and the first two weeks of January, but it still has the potential to affect our results.

In addition to the OLS regressions described above, we also estimate IV regressions in which the December birth dummy is used as an instrument for the dollar value of child-related tax benefits received by December mothers. For analysis of ACS and SIPP data, the set of equations we estimate in carrying out this IV strategy is:

$$\begin{aligned} Y_i &= \sum_{k=1}^{T-1} \alpha_k MonthsElapsed{=}k_i + \sum_{k=1}^{T} \beta_k (MonthsElapsed{=}k \cdot \widehat{TaxValue}_i) + \gamma X_i + \epsilon_i \\ \widehat{TaxValue}_i &= \delta DecBirth_i + \lambda X_i + \nu_i \end{aligned} \tag{3}$$

For IV analysis of Decennial Census data, we replace the *DecBirth* dummy in Equation 2 with the predicted tax savings associated with a December birth. We use the IV estimation strategy with the same set of dependent variables described above. For mothers with children born in December, the *TaxValue* term is defined as the difference between the tax liabilities computed with and without the newborn child, measured in thousands of dollars. For mothers with children born in January, *TaxValue* is set equal to zero. The coefficient can be interpreted as the impact of receiving an additional $1000 in tax savings.

The receipt of child-related tax benefits is not universal, as it requires filing a tax return. We would not expect any difference in the after-tax incomes of December and January mothers who are not filing tax returns. While it would be sensible to estimate regressions

only for the set of people who file returns, or to compare results for filers and non-filers as a further test of whether any labor supply response is causal, we do not observe tax filing status in any of our datasets. However, we expect that the large majority of individuals in our analysis are tax filers. Nationally, about 87% of individuals are represented on a tax return (Orszag and Hall 2003). Filing a return does not necessarily imply claiming all of the child-related benefits to which one is entitled. For example, Plueger (2009) estimates that about 75% of all EITC-eligible individuals both file and receive the credit, and that an additional 9% of EITC-eligible individuals file but do not claim the credit. If the take-up of the EITC is lower than take-up of the dependent exemption, our calculation of child-related tax benefits may be more accurate for high-income filers (who primarily benefit from the dependent exemption) while overstating actual benefits for low-income filers. This caveat should be kept in mind when we later compare results across demographic groups.

5 Results

5.1 Current Employment Status

Table 3 shows the results of regressions predicting whether a mother is employed and currently working at the time of the ACS interview. Results from a reduced-form OLS regression are shown in the first two columns of the table and results from an IV regression are shown in columns 3 and 4. Columns 1 and 3 show the coefficients on the series of *MonthsElapsed* dummy variables measuring the number of months between when a mother gave birth and when she was interviewed. As expected, these coefficients indicate a strong temporal pattern in labor supply. Relative to the omitted group of mothers interviewed in the 12th month after having given birth, women interviewed one or two months after birth are about 30 percentage points less likely to be currently at work. The probability of being currently at work gradually rises over time. It is 8.7 percentage points below the average for the omitted group in the third month after birth, and then mostly returns to a steady long-run level.

These patterns are the same in the OLS and IV regressions. They are also consistent with patterns documented for other samples of new mothers (Laughlin 2011).

The coefficients of particular interest in Table 3 are those on the December interaction terms, shown in columns 2 and 4. These results indicate that, in most time periods, the probability that a December mother is currently working is not statistically different from the probability that a January mother is currently working. One exception to this pattern is in the third month after having a child. The OLS regression indicates that, three months after giving birth, December mothers are 6.3 percentage points less likely than January mothers to be currently working. In the IV regression, an additional \$1000 of *TaxSavings* is associated with a 4.9 percentage point reduction in the probability of being at work in the third month after giving birth.[12] The other exception occurs eight months after birth. In this month, the OLS regression shows that, relative to January mothers, December mothers are 5.4 percentage points less likely to be working. The IV regressions show that \$1000 of *TaxSavings* reduces the probability of working by 4.2 percentage points.

Table 4 shows a parallel set of results estimated using SIPP data. The overall pattern of results is quite similar.[13] December mothers are relatively less likely to be working in the third month after birth and in the eighth month after birth. Relative to January mothers, December mothers are about 5.6 percentage points less likely to be employed and working three months after birth and are 6.3 percentage points less likely to be working eight months after giving birth. In this case, the OLS reduced-form effect of a December birth is associated with an effect almost exactly equal in magnitude to the IV effect associated with an additional

[12]It would be interesting to study additional graphical evidence related to the probability that a mother is working three months after giving birth. In particular, we would like to produce a regression discontinuity figure plotting the share of mothers at work three months after birth, by exact date of birth. The need to protect confidentiality of survey respondents prevents us from producing such a figure. When cells are defined by a combination of exact date of birth and time elapsed between giving birth and the ACS interview, the number of cell-specific observations is often below the Census Bureau's threshold for reporting means.

[13]Our analysis is carried out for women ages 15-50. Women at both the low and high ends of this age range may have labor supply behavior that is quite different from women of intermediate ages. In results not shown, we have restricted our SIPP sample to mothers in a more standard range of child-bearing years, 20-40. Results are very similar for this restricted sample. In the third month after birth, December mothers are 5.4 percentage points less likely to be employed, and this coefficient is statistically significant at the 10% level. Full results are available upon request.

$1000 of child-related tax benefits. This is to be expected, as the average tax benefit received by December mothers in the SIPP sample is very close to $1000.

In the next set of regression results, shown in Table 5, the dependent variable is a dummy equal to one if a woman is employed and is temporarily not working at the time of her ACS interview. Mothers who are on maternity leave when interviewed would fall into this category.[14] The number of observations is smaller here because the sample is restricted to women who are employed at the time of the ACS interview. Being employed and temporarily not working is a relatively uncommon employment status, particularly as time elapsed since giving birth increases. To avoid creating cells containing only a small number of observations, the omitted comparison category is defined differently in this regression. Here, the omitted category includes mothers observed seven to 12 months after having given birth.

Again there is evidence of a strong temporal pattern in employment status around the time of birth. Relative to the omitted group of women interviewed seven to twelve months after giving birth, women observed within one month of giving birth are 54.3 percentage points more likely to be temporarily not working. Temporary absences remain high in the second month after birth and then drop off quickly. The likelihood of temporary absence is statistically indistinguishable from the long-run level by six months after birth. Comparing mothers who give birth in December and in January, in most months there is no statistically significant difference in the probability of being temporarily absent from work. One exception occurs three months after birth. In this month, OLS results indicate that December mothers are 7.0 percentage points more likely to be temporarily not working and IV results indicate that $1000 of tax benefits increases the probability of temporary absence by 5.2 percentage points. The positive coefficients here are consistent with an income effect of the expected direction. The higher levels of after-tax income enjoyed by December mothers are associated with higher probabilities of being absent from work. There is also evidence that December

[14]These mothers may be on either paid or unpaid leave. The employment status variables in the SIPP do not allow identification of a corresponding group. In the SIPP, anyone who is receiving pay from a job is categorized as being employed. It is impossible to identify those who are currently being paid while temporarily absent from work.

mothers are more likely to be absent from work five months after birth, with point estimates about half the size of the effect observed three months after birth.

There are at least two reasons why it is plausible that an income effect on maternal labor supply is consistently observed three months after birth. First, this is a time period in which the 12 weeks of unpaid leave mandated by FMLA run out. We hypothesize that while a woman still has some legal protection that a job will be available to her after maternity leave, additional income may induce her to lengthen her leave. Once that legal protection has expired, in the fourth month after birth and beyond, her labor supply decisions may respond differently to additional income. Second, tax benefits may have the greatest impact on labor supply at the moment that they are received by filers. Three months after birth corresponds to the calendar month of March for December mothers and to April for January mothers. This is the time when many tax refund payments are distributed. Approximately half of all refund dollars are disbursed in March and April combined (LaLumia 2013).

The size of the third-month effect on labor supply, that an additional $1000 of tax benefits reduces the probability of working by approximately 5 percentage points, may seem large in comparison to our results related to birth timing. If relatively few women are manipulating birth timing in response to tax benefits, why are women's labor supply decisions seemingly more sensitive to tax benefits? We offer several potential explanations. First, women may be concerned about negatively affecting the health of their infants through scheduling an early birth (Schulkind and Shapiro 2014, Borra, Gonzalez and Sevilla-Sanz 2015). The choice of when to return to work does not involve the same set of potential health implications. Second, the shifting of a birth generally requires the cooperation of a doctor. Responding to increased after-tax income by delaying the return to work may require the cooperation of an employer. However, particularly in the 12 weeks after birth for women in FMLA-covered employment, this is a choice that is likely more directly in the hands of the mother. Third, tax-motivated birth shifting requires that a mother understand the December 31 discontinuity in tax benefits for children before she actually receives the benefits. Once

tax returns have been filed and refunds received, December and January mothers will have realized different tax treatment even if they do not fully comprehend why.

Evidence of an income effect on labor supply is also consistently present in the eighth month after birth. It is more difficult to find a compelling explanation for why additional income reduces maternal labor supply in this month, which corresponds to August for December mothers. This is a common time for people to travel for vacation (Hong and Yu 2009), and it may be the case that families with December births save the larger tax refunds received in the spring to fund vacations at the end of the summer.[15]

The differing tax treatment of mothers who give birth in December and January does not persist beyond the first year of a child's life.[16] Thus, if the results shown so far indeed represent an income effect on labor supply rather than strategic timing of birth or some other seasonal pattern that differs for December and January mothers, labor supply differences should not persist beyond the first year following a child's birth. To investigate this possibility, we consider mothers interviewed in the ACS anywhere between one and two years after their most recent birth. We define a set of *MonthsElapsed* dummies for these mothers corresponding to the time between a child's first birthday and the interview. We interact these dummies with an indicator for giving birth in December. We estimate regressions predicting whether a mother is currently at work, with the number of months elapsed since the child's first birthday replacing the number of months elapsed since birth. Because December mothers and January mothers in this sample face exactly the same tax treatment, we expect none of the interaction terms to be statistically different from zero. In results not shown, we find that almost all of the coefficients on the *MonthsElapsed* dummies are insignificant.

[15]It is also possible that some portion of the difference between the labor supply of December and January mothers that is observed in the third month after birth is due to vacation timing. The third month after birth, March for December mothers and April for January mothers, can include weeks of spring break school vacations.

[16]Tax treatment does differ again when children age out of eligibility for child-related benefits. For most tax provisions, this occurs when the child turns 18 or when a child who is a full-time student turns 24. Several authors have used variation in the time at which a child becomes ineligible for benefits to estimate labor supply and other effects. See Looney and Singhal (2006), Dokko (2008), and Feldman et al. (2014).

In addition, almost all of the *DecemberBirth* interaction terms are small and insignificant.[17] In particular, there is no divergence in the labor supply of December and January mothers three months after a child's first birthday. This makes it less plausible that the earlier results, estimated for mothers observed up to one year after a child's birth, are merely a reflection of bias due to sample selection. In other words, it is not the case that December mothers are always less likely to work in March than January mothers are to work in April. This weighs against the possibility that December and January mothers systematically work in different industries with different seasonal intensity of labor demand.

Not all mothers face the same constraints in deciding when to return to work after a child's birth. The possibility of relying on the income of their husbands may allow married women to take longer maternity leaves than single women, and may be associated with an increased elasticity of labor supply. Using ACS data, we estimate OLS regressions predicting the probability of being currently working separately for married and for unmarried women. The results are shown in Table 6. For both groups, labor supply falls sharply in the months immediately after birth. Trends in labor supply are similar for December and January mothers in most months, with an exception in the third month after giving birth. The child-related tax benefits associated with giving birth in December reduce the probability of working in the third month after birth by 7.9 percentage points among married women, but have no significant effect among unmarried women. It is worth noting that the sample size is smaller for the unmarried group, and standard errors are larger. With the sample split on the basis of marital status, there is no longer a statistically significant difference in labor supply of December vs. January mothers in the eighth month after birth.

Employers of highly-skilled women may offer greater flexibility in the choice of when to return to a job than employers in unskilled labor markets. If so, income effects on labor supply may be larger for women with higher levels of education. To investigate this

[17] In periods one to eleven months after a child's first birthday, the difference between the probability of being at work for December mothers relative to January mothers is never greater than 1.9 percentage points, and is never significant at the 10% level. Twelve months after a child's first birthday, December mothers are 3.4 percentage points more likely to be at work, significant at the 5% level.

possibility, we divide the sample into groups of mothers with and without some college education, and estimate OLS regressions predicting whether a woman is currently working separately for each educational group. The results are shown in Table 7. Both educational groups display declines in the probability of working in the months just after giving birth. In most months, there is no significant difference in the probability of working across December and January mothers. Again, an exception occurs three months after birth when December mothers with at least some college education are 7.3 percentage points less likely to be at work. Among mothers with no college education, there is no significant difference in the probability that December and January mothers are at work. The smaller size of the low-education group results in larger standard errors, though, and we cannot reject an effect of equal magnitude for low- and high-education groups. The results in Tables 6 and 7 are consistent with Han et al. (2009), who find that legislated increases in available parental leave led to increased use of leave only among married mothers and highly-educated mothers.

While we have argued that being married and having higher educational attainment may provide women with more flexibility in their choices regarding when to return to work after the birth of a child, both marriage and educational attainment are also associated with having higher income. High- and low-income women may have different preferences about the trade-offs associated with prolonging maternity leave. Data from the SIPP allow more direct examination of the potential role for flexible work schedules. For all of the mothers in our SIPP sample, we observe usual hours of work and whether a mother works part-time. Grouping individuals by occupation and using only pre-birth observations, we calculate the standard deviation of hours worked in each occupation. We characterize occupations with above-median standard deviations as providing relatively more flexibility in work schedules. Similarly, we characterize occupations in which a high fraction of workers are part-time as more flexible occupations. Tables 8 and 9 show the results of regressions predicting whether a mother is currently working separately for those in high-flexibility and low-flexibility occupations. Only in the occupations with more flexible work schedules do we see evidence

that December mothers are slower to return to work than January mothers. In these flexible occupations, December mothers are less likely to be at work in each of the first three months after giving birth. In occupations with less flexible schedules, there is no difference in the probabilities that December and January mothers are working in any of the observed months.

5.2 Earned Income

The next set of regressions makes use of a different dependent variable, earned income. In the ACS, income from wages and salary is reported for the twelve months prior to the interview. This is a summary measure of labor supply decisions made throughout the year.[18] If December mothers are slower to return to work because of the tax treatment they receive, then they will have lower amounts of wage income in the year following birth. We do not find evidence of this behavior in the ACS sample. Results are shown in Table 10, with OLS results in the first two columns and IV results in columns 3 and 4. The format of this table matches the format of earlier employment status tables. Columns 1 and 3 report coefficients on a series of dummies measuring the number of months elapsed between a woman's most recent birth and the time at which she is reporting wage income. As expected, women who are reporting wage income just a few months after giving birth report higher values than the comparison group, women interviewed 12 months after giving birth. This is consistent with most of the reference period for their wage reports coming from months before giving birth. The observed temporal pattern in reported wage income is very similar in OLS and IV regressions. Columns 2 and 4 show the differential effects for December mothers. None of the interaction terms is statistically different from zero, indicating that in no month is there a statistically significant difference in reported earnings for the preceding 12-month period. Moreover, the coefficients vary substantially from month to month in columns 2 and

[18] In principle, we could use a measure of hours worked to consider the intensive margin of labor supply. In practice, ACS respondents report the usual number of hours worked per week, averaging over the twelve months prior to the ACS interview. This makes the ACS hours worked variable a poor candidate for analyzing short-run changes in work intensity.

4 despite the large overlap in recall periods from one month to the next.

Results of monthly earnings regressions estimated with SIPP data are shown in Table 11. The dependent variable is the difference between a mother's earned income in a particular month and her average monthly earnings from the six months prior to giving birth. Columns 1 and 3 show the average temporal pattern in earnings. As expected, monthly earnings decline the most shortly after birth. Two months after birth, monthly earnings are about $200 lower than pre-birth levels. Monthly earnings gradually return to pre-birth averages by the end of a year. The temporal pattern is essentially the same in OLS and IV regressions. Columns 2 and 4 present evidence on whether the earnings of December mothers evolve differently. There are several months in which December mothers earn significantly less. December mothers earn less four, five, eight, and ten months after giving birth. Each interaction term shows the differential earnings of December mothers in a particular month. Summing all coefficients in column 2 to yield an annual effect shows that December mothers earn about $988 less than January mothers over the 12 months following birth. Relative to average 12-month earnings, this represents a decline of approximately 6%.[19] In the IV regression, the coefficients in column 4 show the difference in monthly earnings associated with receiving an additional $1 of tax benefit. Again, tax benefits are associated with significant reductions in earnings in several months. Summing all monthly interaction terms indicates that, across a year, each additional dollar of tax benefit reduces maternal earnings by $1.02. It is worth noting that in Table 11 the magnitude of coefficients varies quite a bit from month to month. Our view is that this is largely due to noise inherent in a fairly small sample. To mitigate this problem somewhat, Table 12 aggregates monthly earning amounts up to quarterly earnings. In this specification, the income effects of additional after-tax income on earnings are concentrated in the second quarter after birth. Once again, summing the interaction coefficients to arrive at an annual income effect suggests that each additional

[19]To compute 12-month earnings, we scale up average monthly labor income of $1304. This allows us to use data from all mothers in the sample, including those mothers who are in the SIPP survey for less than a year before giving birth.

dollar of tax benefit reduces labor income by approximately one dollar.

Table 13 shows the results of estimating OLS and IV regressions predicting annual wage income using Decennial Census data. The dependent variable is equal to a woman's total wage and salary income earned during calendar year 1999. We restrict the sample to mothers whose most recent child was born in the last two weeks of December 1998 or in the first two weeks of January 1999. If in fact more generous tax benefits have a negative effect on labor supply, it is for this group of mothers that we would expect to see a negative coefficient on *DecBirth* in the OLS regression, and a negative coefficient on $\widehat{TaxValue}$ in the IV regression. However, these coefficients are statistically not different from zero. The point estimates are positive, which could reflect the mechanical bias due to calendar year aggregation discussed earlier. It is also possible that measurement error is attenuating the results. Discrepancies between reported and actual earnings may be more common when respondents are asked to recall changes over the longer window of time involved in the Decennial Census than in the SIPP. We have also carried out a regression discontinuity analysis of the annual earnings reported by mothers giving birth in December 1998 and in January 1999. An income effect of child-related tax benefits on labor supply would produce discontinuously lower earnings for the December mothers than for the January mothers. Figure 3 shows the results of this analysis. Consistent with the regression results of Table 13, although not with our hypothesis, December mothers report higher average earnings.

6 Conclusion

This paper uses a discontinuity in the tax treatment of children to identify income effects on maternal labor supply. Mothers who give birth at the end of December can claim tax benefits for their children almost a year earlier than mothers who give birth in early January. As a result, in the first year of a child's life, December mothers have higher after-tax income than similar January mothers. However, both December and January mothers face the same

set of marginal tax rates on income earned in the year following the birth. This situation provides a useful opportunity to isolate income effects on labor supply.

The results indicate that mothers who give birth in late December are slower to return to work and earn less in the year following a child's birth. Results from both the ACS and the SIPP indicate that, relative to early January mothers, late December mothers are about six percentage points less likely to be working in the third month after a child's birth. This pattern is evident for married mothers but not for unmarried mothers, and only for mothers with some college education. Analysis of earned income using SIPP data indicates that, aggregated over the year following a child's birth, each additional dollar of child-related tax benefits lowers earnings by approximately one dollar. Estimates from the ACS and Decennial Census show earnings results not statistically different from zero, but differences in the time period over which earnings are reported likely introduce more noise into the ACS and Decennial Census earnings measures than into the SIPP measure.

These estimates are in contrast to the consensus view that income effects on labor supply are quite close to zero. We hypothesize that the income elasticity of labor supply is not a constant, and that labor supply decisions made shortly after the birth of a child are indeed much more income elastic than labor supply decisions made at other points in the lifecycle.

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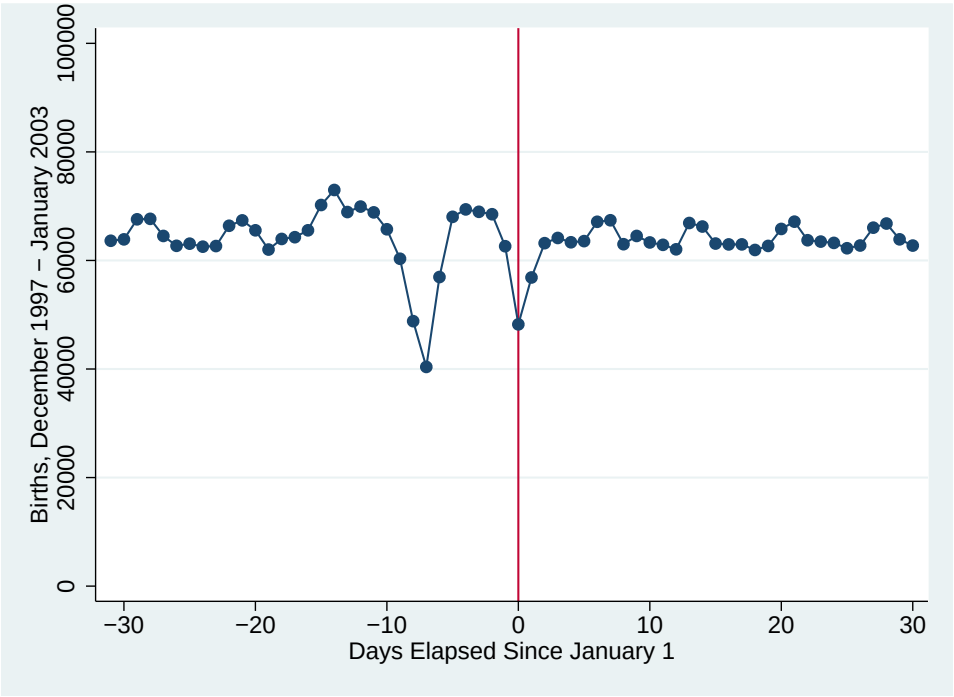
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Figure 1: Total Number of Births Per Day, Matched December-January Pairs from December 1997 to January 2003



Birth counts by exact date of birth were collected from the National Center for Vital Statistics. Each point in the figure aggregates all births occurring on December dates from 1997 to 2002, and on January dates from 1998 to 2003.

Figure 2: Test of Identifying Assumption: Average Annual 1999 Wage Income of Mothers Giving Birth In December 1999 and January 2000



The figure plots average wage income for calendar year 1999, as reported in the 2000 Decennial Census. Each point represents average earnings for all mothers giving birth on a particular day. Mothers giving birth in December 1999 and January 2000 are included.

Figure 3: Average Annual 1999 Wage Income of Mothers Giving Birth in December 1998 and January 1999



The figure plots average wage income for calendar year 1999, as reported in the 2000 Decennial Census. Each point represents average earnings for all mothers giving birth on a particular day. Mothers giving birth in December 1998 and January 1999 are included.

Table 1: Summary Statistics, December and January Mothers

	Full Sample	January Births	December Births
A. ACS Sample			
Age	28.85	28.81	28.88
	(5.94)	(5.95)	(5.94)
White	0.73	0.74	0.73
	(0.41)	(0.41)	(0.42)
Married	0.71	0.71	0.71
	(0.43)	(0.44)	(0.43)
Has Male Spouse/Partner	0.80	0.80	0.80
	(0.38)	(0.38)	(0.38)
Number of Kids<19	1.96	1.95	1.96
	(1.03)	(1.03)	(1.03)
Some College	0.21	0.21	0.20
	(0.39)	(0.40)	(0.39)
College Grad	0.44	0.43	0.45
	(0.50)	(0.50)	(0.50)
Child Tax Benefit, Actual Income	1205.69	1200.50	1210.93
	(776.94)	(777.03)	(776.83)
Child Tax Benefit, Predicted Income	1300.65	1296.22	1305.12
	(761.65)	(763.44)	(759.92)
N of observations	16384	8158	8226
B. SIPP Sample			
Age	28.52	28.48	28.56
	(5.92)	(5.93)	(5.92)
White	0.81	0.83	0.79
	(0.39)	(0.37)	(0.41)
Married	0.74	0.76	0.72
	(0.44)	(0.43)	(0.45)
N of Dependents	1.98	1.94	2.02
	(0.99)	(0.98)	(1.01)
Some College	0.33	0.34	0.32
	(0.47)	(0.47)	(0.47)
College Grad	0.30	0.29	0.31
	(0.46)	(0.46)	(0.46)
Labor Earnings	1303.99	1310.55	1297.36
	(2215.76)	(2216.90)	(2216.71)
Child Tax Benefit, Pre-Birth Income	978.09	979.67	976.50
	(636.82)	(626.33)	(647.83)
N of Mothers	1051	523	528

The table reports means, with standard deviations in parentheses.

Table 2: Predicting Whether Child Is Born in December, ACS Sample

	Full Sample	Two-Week Sample
Child Tax Benefits ($1000s)	0.005	0.017**
	(0.006)	(0.007)
Age	-0.001	0.008
	(0.005)	(0.008)
Age Squared	0.00001	0.0001
	(0.00008)	(0.0001)
White	-0.012	-0.008
	(0.010)	(0.013)
Some College	-0.006	-0.021
	(0.009)	(0.016)
College Grad	0.0003	0.002
	(0.009)	(0.014)
Married	0.013	0.016
	(0.010)	(0.012)
First Child	-0.015	-0.020
	(0.010)	(0.012)
Spouse/Partner Income ($1000s)	0.00002	0.00007
	(0.00006)	(0.0001)
Constant	0.443**	0.314***
	(0.101)	(0.159)
N	16384	7932

The table reports coefficients and standard errors from unweighted OLS regressions predicting whether a child is born in December. Each regression also includes year fixed effects, state fixed effects, and day-of-week fixed effects. Standard errors are clustered at the state level.

Table 3: Predicting Whether Mom is Currently Employed and Working, ACS Results

	OLS Results		IV Results	
	Months Elapsed Since Birth	Months Interacted with DecBirth	Months Elapsed Since Birth	Months Interacted with $\widehat{TaxValue}$
	(1)	(2)	(3)	(4)
Month 1	-0.331***	0.019	-0.331***	0.016
	(0.031)	(0.033)	(0.031)	(0.026)
Month 2	-0.305***	0.015	-0.305***	0.013
	(0.027)	(0.023)	(0.027)	(0.019)
Month 3	-0.087***	-0.063**	-0.087***	-0.049**
	(0.031)	(0.030)	(0.031)	(0.023)
Month 4	-0.049	0.011	-0.049	0.009
	(0.031)	(0.034)	(0.031)	(0.026)
Month 5	-0.002	-0.034	-0.002	-0.026
	(0.029)	(0.021)	(0.029)	(0.016)
Month 6	0.002	0.008	0.002	0.007
	(0.017)	(0.024)	(0.017)	(0.020)
Month 7	-0.047*	0.019	-0.047*	0.015
	(0.024)	(0.031)	(0.024)	(0.025)
Month 8	-0.002	-0.054**	-0.002	-0.042**
	(0.027)	(0.024)	(0.027)	(0.019)
Month 9	0.007	-0.023	0.007	-0.018
	(0.023)	(0.024)	(0.023)	(0.020)
Month 10	0.016	0.017	0.016	0.013
	(0.026)	(0.028)	(0.026)	(0.022)
Month 11	0.002	0.026	0.002	0.021
	(0.023)	(0.026)	(0.023)	(0.021)
Month 12		-0.001		-0.001
		(0.027)		(0.021)
N	16384		16384	

Columns 1 and 2 report unweighted OLS coefficients and standard errors. Columns 3 and 4 report second-stage results from an IV regression. Each regression also controls for maternal age and age squared, income earned by a male spouse or partner (set equal to zero if there is no male partner), the number of own under-19 children in the household, state fixed effects, year fixed effects (where a year is defined as an adjacent December/January pair), the number of days elapsed between December 1 and the birth, day-of-week dummies for the date of birth, and dummies for being white, having some college education, having completed a college degree, and being married.

Table 4: Predicting Whether Mom is Employed and Working in Current Month, SIPP Results

	OLS Results		IV Results	
	Months Elapsed Since Birth	Months Interacted with DecBirth	Months Elapsed Since Birth	Months Interacted with $\widehat{TaxValue}$
	(1)	(2)	(3)	(4)
Month 1	-0.045*	-0.025	-0.045*	-0.025
	(0.027)	(0.025)	(0.027)	(0.025)
Month 2	-0.068***	-0.041	-0.069***	-0.042
	(0.026)	(0.026)	(0.026)	(0.026)
Month 3	-0.059**	-0.056**	-0.060**	-0.057**
	(0.025)	(0.028)	(0.025)	(0.029)
Month 4	-0.048**	-0.025	-0.048**	-0.026
	(0.024)	(0.027)	(0.024)	(0.028)
Month 5	-0.024	-0.026	-0.025	-0.027
	(0.022)	(0.027)	(0.022)	(0.028)
Month 6	-0.015	-0.040	-0.015	-0.041
	(0.021)	(0.027)	(0.021)	(0.028)
Month 7	-0.017	-0.035	-0.018	-0.037
	(0.020)	(0.027)	(0.020)	(0.029)
Month 8	-0.013	-0.063**	-0.013	-0.065**
	(0.019)	(0.028)	(0.019)	(0.029)
Month 9	-0.019	-0.032	-0.020	-0.032
	(0.018)	(0.029)	(0.018)	(0.029)
Month 10	-0.016	-0.014	-0.016	-0.015
	(0.015)	(0.029)	(0.015)	(0.030)
Month 11	-0.000	-0.014	-0.001	-0.014
	(0.013)	(0.030)	(0.013)	(0.031)
Month 12		-0.024		-0.025
		(0.031)		(0.032)
N	11,075		11,075	
R-squared	0.26			

Columns 1 and 2 report unweighted OLS coefficients. Columns 3 and 4 report second-stage results from an IV regression. Reported standard errors are clustered at the mother level. Each regression also controls for pre-birth monthly earnings, maternal age and age squared, income earned by a husband (set equal to zero if there is no spouse), the number of own children in the household, state fixed effects, year fixed effects (where a year is defined as an adjacent December/January pair), and dummies for being white, having some college education, having completed a college degree, and being married.

Table 5: Predicting Whether Mom is Employed and Temporarily Not Working, ACS Results

	OLS Results		IV Results	
	Months Elapsed Since Birth	Months Interacted with DecBirth	Months Elapsed Since Birth	Months Interacted with $\widehat{TaxValue}$
	(1)	(2)	(3)	(4)
Month 1	0.543***	-0.014	0.543***	-0.010
	(0.030)	(0.047)	(0.030)	(0.034)
Month 2	0.477***	-0.002	0.477***	-0.002
	(0.024)	(0.028)	(0.024)	(0.021)
Month 3	0.151***	0.070**	0.151***	0.052**
	(0.021)	(0.027)	(0.021)	(0.020)
Month 4	0.067***	-0.0001	0.067***	0.0002
	(0.021)	(0.026)	(0.021)	(0.019)
Month 5	0.029*	0.034*	0.029***	0.025*
	(0.011)	(0.019)	(0.011)	(0.014)
Month 6	0.017	0.018	0.017	0.014
	(0.011)	(0.017)	(0.011)	(0.014)
Months 7-12		0.021		0.015
		(0.012)		(0.009)
N	9863		9863	

Columns 1 and 2 report unweighted OLS coefficients and standard errors. Columns 3 and 4 report second-stage results from an IV regression. Each regression also controls for maternal age and age squared, income earned by a male spouse or partner (set equal to zero if there is no male partner), the number of own under-19 children in the household, state fixed effects, year fixed effects (where a year is defined as an adjacent December/January pair), the number of days elapsed between December 1 and the birth, day-of-week dummies for the date of birth, and dummies for being white, having some college education, having completed a college degree, and being married.

Table 6: Heterogeneity in Probability of Being Currently Employed and Working by Marital Status, ACS Results

	Mother is Married		Mother is Not Married	
	Months Elapsed Since Birth	Months Interacted with DecBirth	Months Elapsed Since Birth	Months Interacted with DecBirth
	(1)	(2)	(3)	(4)
Month 1	-0.308***	-0.007	-0.397***	0.107**
	(0.036)	(0.041)	(0.041)	(0.046)
Month 2	-0.295***	0.008	-0.340***	0.051
	(0.032)	(0.031)	(0.041)	(0.042)
Month 3	-0.089**	-0.079**	-0.074**	-0.008
	(0.036)	(0.037)	(0.043)	(0.055)
Month 4	-0.041	0.007	-0.071	0.027
	(0.037)	(0.038)	(0.051)	(0.056)
Month 5	-0.010	-0.026	0.018	-0.045
	(0.029)	(0.028)	(0.049)	(0.059)
Month 6	0.014	0.005	-0.030	0.013
	(0.019)	(0.033)	(0.046)	(0.055)
Month 7	-0.061**	0.015	0.001	0.040
	(0.030)	(0.041)	(0.044)	(0.056)
Month 8	-0.021	-0.044	0.044	-0.070
	(0.028)	(0.029)	(0.050)	(0.059)
Month 9	0.009	-0.021	-0.002	-0.016
	(0.024)	(0.032)	(0.043)	(0.045)
Month 10	0.011	0.018	0.033	0.018
	(0.033)	(0.030)	(0.047)	(0.052)
Month 11	-0.015	0.047	0.036	-0.017
	(0.030)	(0.031)	(0.040)	(0.051)
Month 12		-0.004		0.025
		(0.029)		(0.055)
N	12270		4114	

The table reports coefficients and standard errors from unweighted OLS regressions. The additional controls not shown in this table are maternal age and age squared, income earned by a male spouse or partner (set equal to zero if there is no male partner), the number of own under-19 children in the household, state fixed effects, year fixed effects (where a year is defined as an adjacent December/January pair), the number of days elapsed between December 1 and the birth, day-of-week dummies for the date of birth, a dummy for being white, and dummies for having some college education or being a college graduate.

Table 7: Heterogeneity in Probability of Being Currently Employed and Working by Educational Attainment, ACS Results

	Some College Education		No College Education	
	Months Elapsed Since Birth	Months Interacted with DecBirth	Months Elapsed Since Birth	Months Interacted with DecBirth
	(1)	(2)	(3)	(4)
Month 1	-0.329***	0.044	-0.345***	-0.035
	(0.039)	(0.042)	(0.046)	(0.042)
Month 2	-0.304***	0.008	-0.310***	0.034
	(0.036)	(0.027)	(0.031)	(0.048)
Month 3	-0.081**	-0.073**	-0.107**	-0.052
	(0.037)	(0.036)	(0.044)	(0.067)
Month 4	-0.053	0.020	-0.050	-0.008
	(0.037)	(0.037)	(0.047)	(0.055)
Month 5	0.011	-0.035	-0.041	-0.022
	(0.031)	(0.025)	(0.046)	(0.057)
Month 6	0.013	0.015	-0.033	-0.003
	(0.023)	(0.030)	(0.056)	(0.064)
Month 7	-0.033	0.011	-0.093*	0.041
	(0.036)	(0.040)	(0.048)	(0.051)
Month 8	0.002	-0.044*	-0.016	-0.069
	(0.028)	(0.025)	(0.046)	(0.060)
Month 9	0.033	-0.050*	-0.062	0.043
	(0.029)	(0.028)	(0.053)	(0.052)
Month 10	0.039	0.003	-0.050	0.061
	(0.033)	(0.032)	(0.050)	(0.069)
Month 11	0.009	0.031	-0.022	0.010
	(0.033)	(0.036)	(0.037)	(0.052)
Month 12		0.003		-0.012
		(0.032)		(0.059)
N	11812		4572	

The table reports coefficients and standard errors from unweighted OLS regressions. The additional controls not shown in this table are maternal age and age squared, income earned by a male spouse or partner (set equal to zero if there is no male partner), the number of own under-19 children in the household, state fixed effects, year fixed effects (where a year is defined as an adjacent December/January pair), the number of days elapsed between December 1 and the birth, day-of-week dummies for the date of birth, a dummy for being white, and a dummy for being married.

Table 8: Heterogeneity in Probability of Being Currently Employed and Working by Degree of Job Flexibility (sd hours worked), SIPP Results

	High sd hours worked		Low sd hours worked	
	Months Elapsed Since Birth	Months Interacted with DecBirth	Months Elapsed Since Birth	Months Interacted with DecBirth
	(1)	(2)	(3)	(4)
Month 1	-0.071**	-0.105***	-0.144***	0.063*
	(0.035)	(0.039)	(0.034)	(0.036)
Month 2	-0.108***	-0.083**	-0.141***	0.010
	(0.038)	(0.040)	(0.033)	(0.039)
Month 3	-0.103***	-0.076*	-0.097***	-0.022
	(0.038)	(0.043)	(0.031)	(0.040)
Month 4	-0.056	-0.012	-0.072**	0.007
	(0.035)	(0.037)	(0.028)	(0.036)
Month 5	-0.004	-0.028	-0.048*	0.021
	(0.034)	(0.032)	(0.029)	(0.032)
Month 6	-0.011	-0.005	-0.015	-0.024
	(0.033)	(0.031)	(0.027)	(0.031)
Month 7	-0.002	-0.028	-0.024	0.019
	(0.032)	(0.032)	(0.027)	(0.027)
Month 8	0.003	-0.083**	-0.004	-0.013
	(0.030)	(0.036)	(0.023)	(0.028)
Month 9	0.001	-0.065*	-0.030	0.024
	(0.029)	(0.036)	(0.026)	(0.028)
Month 10	0.011	-0.034	-0.028	0.026
	(0.030)	(0.033)	(0.022)	(0.029)
Month 11	0.019	-0.009	-0.007	0.019
	(0.025)	(0.032)	(0.019)	(0.025)
Month 12		0.028		-0.018
		(0.033)		(0.028)
N	3909		3988	

The table reports coefficients and standard errors from unweighted OLS regressions. Columns (1) and (2) include occupations with standard deviation of hours worked above above the sample median. Columns (3) and (4) include occupations below the sample median. See text for details. Reported standard errors are clustered at the mother level. Each regression also controls for pre-birth monthly earnings, maternal age and age squared, income earned by a male spouse or partner (set equal to zero if there is no male partner), the number of own children in the household, state fixed effects, year fixed effects (where a year is defined as an adjacent December/January pair), and dummies for being white, having some college education, having completed a college degree, and being married.

Table 9: Heterogeneity in Probability of Being Currently Employed and Working by Degree of Job Flexibility (share of part time work), SIPP Results

	High share part time		Low share part time	
	Months Elapsed Since Birth	Months Interacted with DecBirth	Months Elapsed Since Birth	Months Interacted with DecBirth
	(1)	(2)	(3)	(4)
Month 1	-0.108***	-0.104**	-0.105***	0.041
	(0.039)	(0.043)	(0.031)	(0.032)
Month 2	-0.138***	-0.093**	-0.111***	0.001
	(0.039)	(0.044)	(0.031)	(0.035)
Month 3	-0.119***	-0.079*	-0.077***	-0.041
	(0.039)	(0.045)	(0.028)	(0.036)
Month 4	-0.081**	-0.008	-0.046*	-0.012
	(0.037)	(0.042)	(0.026)	(0.030)
Month 5	-0.003	-0.036	-0.045	0.020
	(0.035)	(0.036)	(0.027)	(0.028)
Month 6	0.010	-0.037	-0.031	0.005
	(0.034)	(0.035)	(0.026)	(0.027)
Month 7	0.033	-0.060*	-0.051*	0.045*
	(0.032)	(0.033)	(0.028)	(0.027)
Month 8	0.033	-0.073**	-0.028	-0.022
	(0.030)	(0.036)	(0.023)	(0.029)
Month 9	-0.006	-0.052	-0.022	0.010
	(0.030)	(0.039)	(0.024)	(0.027)
Month 10	0.010	-0.045	-0.024	0.033
	(0.030)	(0.038)	(0.021)	(0.026)
Month 11	0.016	-0.031	-0.002	0.041**
	(0.024)	(0.036)	(0.019)	(0.021)
Month 12		-0.028		0.036*
		(0.039)		(0.021)
N	3645		4252	

The table reports coefficients and standard errors from unweighted OLS regressions. Columns (1) and (2) include occupations with share of part time workers above above the sample median. Columns (3) and (4) include occupations below the sample median. See text for details. Reported standard errors are clustered at the mother level. Each regression also controls for pre-birth monthly earnings, maternal age and age squared, income earned by a male spouse or partner (set equal to zero if there is no male partner), the number of own children in the household, state fixed effects, year fixed effects (where a year is defined as an adjacent December/January pair), and dummies for being white, having some college education, having completed a college degree, and being married.

Table 10: Predicting Dollar Amount of Annual Wage Income, ACS Results

	OLS Results		IV Results	
	Months Elapsed Since Birth	Interacted with Dec Birth	Months Elapsed Since Birth	Interacted with $\widehat{TaxValue}$
	(1)	(2)	(3)	(4)
Month 1	2784**	122	2784**	0.091
	(1268)	(1076)	(1264)	(0.853)
Month 2	2400**	656	2400**	0.538
	(1005)	(1564)	(1005)	(1.303)
Month 3	3825***	302	3826***	0.233
	(1082)	(1797)	(1082)	(1.418)
Month 4	3878***	-915	3877***	-0.722
	(1218)	(991)	(1215)	(0.776)
Month 5	1362	75	1360	0.053
	(1212)	(988)	(1211)	(0.780)
Month 6	2023*	303	2023*	0.247
	(1174)	(1149)	(1175)	(0.967)
Month 7	1092	698	1091	0.549
	(990)	(1272)	(991)	(1.004)
Month 8	1611	-1262	1609	-0.989
	(1092)	(1175)	(1093)	(0.911)
Month 9	729	1079	727	0.896
	(1190)	(1228)	(1192)	(1.021)
Month 10	1643	-172	1644	-0.138
	(1048)	(1400)	(1047)	(1.092)
Month 11	1156	-2347	1155	-1.874
	(1141)	(1555)	1140	(1.229)
Month 12		-488		-0.382
		(1450)		(1.128)
N	16384		16384	

Columns 1 and 2 report unweighted OLS coefficients and standard errors. Columns 3 and 4 report second-stage results from an IV regression. Each regression also controls for maternal age and age squared, income earned by a male spouse or partner (set equal to zero if there is no male partner), the number of own under-19 children in the household, state fixed effects, year fixed effects (where a year is defined as an adjacent December/January pair), the number of days elapsed between December 1 and the birth, day-of-week dummies for the date of birth, and dummies for being white, having some college education, having completed a college degree, and being married.

Table 11: Predicting Dollar Amount of Monthly Wage Income, SIPP Results

	OLS Results		IV Results	
	Months Elapsed Since Birth	Months Interacted with DecBirth	Months Elapsed Since Birth	Months Interacted with $\widehat{TaxValue}$
	(1)	(2)	(3)	(4)
Month 1	-88.816	-66.390	-90.078	-0.068
	(63.538)	(57.789)	(63.710)	(0.059)
Month 2	-206.765***	-4.374	-207.981***	-0.005
	(59.547)	(59.336)	(59.684)	(0.060)
Month 3	-191.422***	-69.555	-192.575***	-0.072
	(61.389)	(65.128)	(61.398)	(0.067)
Month 4	-123.063**	-108.233*	-124.222**	-0.112*
	(53.534)	(64.126)	(53.554)	(0.066)
Month 5	-68.344	-130.220**	-69.384	-0.135**
	(54.318)	(62.557)	(54.293)	(0.065)
Month 6	-105.436**	-53.970	-106.447**	-0.056
	(51.779)	(65.041)	(51.750)	(0.067)
Month 7	-90.392*	-105.818	-91.125*	-0.111
	(50.553)	(66.208)	(50.460)	(0.069)
Month 8	-97.569**	-114.566*	-98.402**	-0.119*
	(46.934)	(68.201)	(46.839)	(0.070)
Month 9	-104.980**	-84.917	-105.794**	-0.087
	(43.024)	(68.895)	(42.907)	(0.070)
Month 10	-43.380	-119.876*	-43.681	-0.124*
	(39.029)	(72.296)	(38.888)	(0.075)
Month 11	-23.481	-36.329	-23.723	-0.038
	(32.616)	(76.216)	(32.500)	(0.078)
Month 12		-94.123		-0.099
		(79.991)		(0.084)
Total Effect		-988.370		-1.026
		(238.114)		(0.246)
N	11,075		11,075	
R-squared	0.10			

Columns 1 and 2 report unweighted OLS coefficients. Columns 3 and 4 report second-stage results from an IV regression. Reported standard errors are clustered at the mother level. Each regression also controls for pre-birth monthly earnings, maternal age and age squared, income earned by a spouse (set equal to zero if there is no husband), the number of own children in the household, state fixed effects, year fixed effects (where a year is defined as an adjacent December/January pair), and dummies for being white, having some college education, having completed a college degree, and being married.

Table 12: Predicting Dollar Amount of Quarterly Wage Income, SIPP Results

	OLS Results		IV Results	
	Months Elapsed Since Birth	Months Interacted with DecBirth	Months Elapsed Since Birth	Months Interacted with $\widehat{TaxValue}$
	(1)	(2)	(3)	(4)
Quarter 1	-418.315***	-139.585	-421.256***	-0.144
	(153.862)	(160.302)	(154.392)	(0.164)
Quarter 2	-227.873*	-294.034*	-230.445*	-0.305*
	(129.187)	(177.671)	(129.329)	(0.184)
Quarter 3	-223.168**	-305.954	-224.952**	-0.317
	(94.075)	(186.955)	(94.003)	(0.193)
Quarter 4		-250.699		-0.262
		(215.072)		(0.223)
Total Effect		-990.271		-1.028
		(237.786)		(0.246)
N	11,075		11,075	
R-squared	0.10			

Columns 1 and 2 report unweighted OLS coefficients. Columns 3 and 4 report second-stage results from an IV regression. Reported standard errors are clustered at the mother level. Each regression also controls for pre-birth monthly earnings, maternal age and age squared, income earned by a spouse (set equal to zero if there is no husband), the number of own children in the household, state fixed effects, year fixed effects (where a year is defined as an adjacent December/January pair), and dummies for being white, having some college education, having completed a college degree, and being married.

Table 13: Predicting Dollar Amount of Post-Birth Annual Wage Income, Decennial Census Results

	OLS Results	IV Results
Dec Birth	922	
	(650)	
$\widehat{TaxValue}$		894
		(625)
N	33,397	33,397
F-stat from first stage		680.8

Column 1 reports unweighted OLS coefficients and standard errors. Column 2 reports second-stage results from an IV regression. The dependent variable is the amount of wage income earned in calendar year 1999. Each regression also controls for maternal age and age squared, income earned by a male spouse or partner (set equal to zero if there is no male partner), the number of own under-19 children in the household, state fixed effects, the number of days elapsed between December 1 and the birth, day-of-week dummies for the date of birth, and dummies for being white, having some college education, having completed a college degree, and being married. The sample is restricted to mothers giving birth in the last two weeks of December 1998 or the first two weeks of January 1999.